

1 Mathematical Framework

1.1 Model (Prototype)

The model depends on hyperparameters

$$\sigma_1, \dots, \sigma_K, \lambda_1^{(0)}, \dots, \lambda_K^{(0)}, \lambda_1^{(1)}, \dots, \lambda_K^{(1)}, n_1, \dots, n_K, c_1, \dots, c_K$$

$1, 2, \dots, K$ arms, $1, 2, \dots, T$ Time Horizon

$$Y_{i,t}^{(\sigma_i)} \sim N(Z_{i,t}, \sigma_i^2) \text{ Reward of arm } k \text{ at time } t$$

$$Z_{i,t} = \mathbb{E}(Y_{i,t}^{(\sigma)}) \in \mathcal{Z}$$

$$\mathcal{Z} = \left\{ Z \in \{0, 1\}^{K \times T} : \sum_{t=1}^T Z_{i,t} = n_i, \sum_{t=1}^{T-1} Z_{i,t} Z_{i,t+1} \geq c_i \right\}$$

$a_t \subseteq \{1, \dots, K\}$ active arms at time t

Let $1_{i,t} = 1(i \in a_t)$, Let $H_i^{(0)}(t) = \min\{s \geq 1 : 1_{i,t+s} \neq 1_{i,t}, 1_{i,t} = 0\}$ Sleeping holding time

Let $H_i^{(1)}(t) = \min\{s \geq 1 : 1_{i,t+s} \neq 1_{i,t}, 1_{i,t} = 1\}$ Awake holding time

$$H_i^{(j)}(t) \sim \text{Geom}(\lambda_i^{(j)}), \text{ i.e. } \mathbb{P}(H_i^{(j)}(t) = s) = (1 - \lambda_i^{(j)})^{s-1} \lambda_i^{(j)}, \quad s \geq 1$$

Let X_t be the decisions of the algorithm $f \in \mathcal{F}_t$ s.t $\mathcal{F}_t = \left\{ f : \{Y_{X_s,s}^{(\sigma_{X_s})}, X_s\}_{s=1}^t \rightarrow \{1, \dots, K\} \right\} \quad \forall t \in \{1, \dots, T\}$

We now formalize the regret:

$$R_T = \mathbb{E} \left(\sum_{t=1}^T Y_{i^*,t}^{(\sigma_{i^*})} 1_{i^*,t} - \sum_{t=1}^T Y_{X_t,t}^{(\sigma_{X_t})} \right) = \sum_{t=1}^T Z_{i^*,t} 1_{i^*,t} - \sum_{t=1}^T Z_{X_t,t}$$

$$i^* = \arg \max_{i \in [K]} \sum_{t=1}^T Z_{i,t} 1_{i,t}$$

1.2

2 Supplementary Material for Code

- `src/hybrid.py` <class object> `gauss_hybrid_model` : Refer to A1

3 A1

Algorithm 1 Generate Adversarial Sequence

Require: Horizon T , Autocorrelation Parameter c , Observation Ratio ρ

Ensure: Sequence $X_1, X_2, \dots, X_T$ where $X_i \in \{0, 1\}$, $\frac{1}{T} \sum_{t=1}^T X_t = \rho$

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1:  $X_1 \sim \text{Bernoulli}(\rho)$ 
2: for  $t = 2, \dots, T$  do
3:    $\hat{p}_t \leftarrow \frac{1}{t-1} \sum_{s=1}^{t-1} X_s$   $\triangleright$  Running empirical mean
4:    $e_t \leftarrow \hat{p}_t - \rho$   $\triangleright$  Error from target
5:   if  $X_{t-1} = 1$  then
6:      $p_{\text{stay}} \leftarrow \frac{1+c}{2} - e_t$ 
7:   else
8:      $p_{\text{stay}} \leftarrow \frac{1+c}{2} + e_t$ 
9:    $p_{\text{stay}} \leftarrow \text{clip}(p_{\text{stay}}, 0, 1)$ 
10:   $\text{stay} \sim \text{Bernoulli}(p_{\text{stay}})$ 
11:  if  $\text{stay}$  then
12:     $X_t \leftarrow X_{t-1}$ 
13:  else
14:     $X_t \leftarrow 1 - X_{t-1}$ 
15: return  $X_1, \dots, X_T$ 
```
